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Procedia Computer Science 256 (2025) 28–35

Procedia Computer Science

www.elsevier.com/locate/procedia

CENTERIS – International Conference on ENTERprise Information Systems / ProjMAN – International Conference on Project MANagement / HCist – International Conference on Health and Social Care Information Systems and Technologies 2024

Exploratory customer discovery on Veblen goods using ChatGPT and agent-based modeling system for business simulation

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Abstract

The study of demand in a dynamic and complex market economy has been a point of discussion in observing consumer behavior. Entrepreneurs have struggled with market research as customer discovery can be costly and time-consuming. The concept behind non-intuitive demand curves has increased the complexities of predicting demand. *Large language models* (LLMs) are observed to be computational models for humans, having the innate capacity to simulate human behavior and decision-making. Through a simulation model and the *agent-based modeling system* (ABMS) approach, entrepreneurs can utilize LLMs by using their ability for customer discovery. By applying prompting frameworks and *chain-of-thought* (CoT), LLMs such as ChatGPT-3.5 can generate customer *willingness-to-pay* (WTP) without fine-tuning the model. The study experimented on two business models that pertained to Veblen demand and prompted GPT-3.5 to simulate WTP without any explicit references to "Veblen" to determine its inherent knowledge. The results show that some customer simulations generated with GPT-3.5 appear to form a non-inverted demand curve. Future studies may be considered to understand the LLM's capability of perceiving socio-economic and psycho-emotional behavior regarding product demand and consumption.

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Peer-review under responsibility of the scientific committee of the CENTERIS - International Conference on ENTERprise Information Systems / ProjMAN - International Conference on Project MANagement / HCist - International Conference on Health and Social Care Information Systems and Technologies

Keywords: Veblen demand; Customer discovery; Agent-based modeling system; Large language models; ChatGPT; Economic theory; Computer simulation; Economic theory; Simulations; Market validation; Artificial intelligence; AI

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10.1016/j.procs.2025.02.092

1. Introduction

Entrepreneurs who intend to enter an industry may seek relevant information on possible customer demand for their products to strategize their decisions in unfamiliar conditions. However, traditional customer discovery methods such as interviews, surveys, and focus group discussions (FGDs) are costly [10, 18-20]. The entrepreneur is then left with limited options to create decisions in uncertain conditions [17]. *Simulations* have been proposed as an alternative to traditional customer discovery [14], offering low-cost alternatives by operating with a given set of parameters [21]. Exploratory research has been done on using the innate knowledge of *large language models* (LLMs) as the engine behind simulations [14-15]. The *agent-based modeling system* (ABMS) handles interactions and relationships between individual and autonomous agents [23, 34] and their corresponding entrepreneurial and market contexts. In essence, the user provides the attributes given to these agents, sets the environment of the simulation, and allows the elements to interact in a system with a given situation [33]. Along with ABMS, the generated human subjects were experimented on according to different contexts [14, 23]. With the concept of *homo silicus*, LLMs are designed to serve as computational models of humans and are trained to possess latent social information [25]. By utilizing the capabilities of LLMs, entrepreneurs can determine if the generated behavior from the provided business concepts would reflect the phenomenon of Veblen as well.

The law of demand [1] serves as a focal point of discussion in scholarly discussion. It is composed of how people view market prices given their scarce resources [2]. Economists typically model demand for a product as inversely proportional to its price [1, 3]. However, some events show that certain industries can defy the theoretical rule [4]. This "law of demand" is common but not universal. Certain types of goods display non-intuitive demand curves [9]. This paper focuses specifically on "Veblen goods," products for which the demand is directly proportional to its price [5-6]. This characteristic is theorized to be caused by these goods being bought for status rather than for pure utility [7-8]. Automobiles, luxury wines, and high-end watches are examples of Veblen goods [11-13]. Customer *willingness-to-pay* (WTP), in this context, is derived from the idea behind conspicuous consumption as a driver of luxurious appeal [13].

The study explores ChatGPT as a simulation tool for customer discovery, in the case of non-intuitive demand. By inciting its self-learning capabilities, the LLM will be tested on its ability to learn from its outputs and reflections. This can be observed by allowing it to generate customer personas through a set of provided attributes, reflect on the business models, and simulate customer behavior and feedback.

RQ1) What general approach may be taken to determine whether the LLM is inherently knowledgeable of Veblen products and demand, and if it can simulate customer willingness-to-pay (WTP) on certain business models?

RQ2) What are the notable considerations involved in using LLMs to simulate WTP in the context of Veblen or non-intuitive demand curves?

Note that the study does not assert that the simulations done through ChatGPT be considered replacements for real-world customer discovery. This acts as a mere proposal to see if the LLM can serve as an accurate co-pilot to business founders on whether or not it understands such business models and consumer behavior. The first research discussion point aims to identify ChatGPT's capability of *inherently* identifying the category of business models, in this case, Veblen goods. This can determine whether its data training on human behavior suffices for non-intuitive industries. It seeks to understand if the LLM, provided with "some help," or conservative prompt engineering, can generate outputs with better accuracy.

2. Review of related literature

Economic theory suggests that the law of demand follows a complex and dynamic system of relationships [31]. In the case of normal consumption, individuals prefer to purchase cheaper alternatives in a perfectly competitive market [32]. As market demand can dictate the value of products, businesses experience the need to strengthen their pricing decisions so as not to be ruled out by an elastic market [2]. In identifying how the economy functions in relation to the circulation of value and money, economic theory explains such phenomena using statistical analysis, logical reasoning, and models [31]. In [22, 24], demand for normal goods was observed to increase as the real income of the consumer increased. Previously discussed in the theory behind Veblen, this correlates to the higher standard of living from the prestige that product consumption brings to the consumer [12, 26]. Exploring how Veblen demand occurs

consists of identifying specific industries that resemble the factors behind such movement. In the case of non-intuitive demand curves, consumer products such as luxury accessories, clothes, and automobiles [12-13] exploit the Veblen effect and exhibit conspicuous consumption. A research study in Russia [12] observed the growing trend behind car loans as a result of higher income status. It presents how people have become more inclined to incur liabilities to purchase "prestigious" vehicles, connecting to conspicuous consumption. In [26-27], consumers possess the urge to keep in proximity with high-end brands to maintain their social status. Hence, the WTP for luxurious items generally increases as the price increases, although extending beyond *ceteris paribus*, the inclusion of socio-economic status and household income produce complexities with human behavior [11, 13].

In the context of performing customer discovery through ABMS approach, prompting is a required task [14, 16, 25]. No-shot prompting requires the LLM to generate output based solely on the task description, refraining from endowing it with any examples. No-shot prompting is dependent on specific situations, and requires the prompt to be explicit, concrete, and complete on its request in order to avoid potential hallucinations [28-29]. Another recent study [14] revealed that the zero-shot learning approach serves as a useful tool for entrepreneurs to determine the potential WTP of consumers targeted by their businesses. Through the implementation of *chain-of-thought* (CoT) techniques, the entrepreneur does not have to fine-tune the LLM, therefore avoiding larger costs in doing customer discovery [35]. By providing the agents [23] with certain preferences, biases, economic, and social information, their behavior can be explored through simulations. A study [25] conducted using a hiring scenario illustrated the ability of the LLM to replicate employee responses to wage or compensation rates. The user can utilize *homo silicus* to predict social interactions, but only to a certain extent [30]. Although the model is pre-trained on large amounts of data, it cannot immediately understand the underlying conditions and factors that human subjects perceive. In [14], a CoT prompting framework was presented, generating customer WTP to a tutoring service business which resembles the general law of demand in the context provided. However, this does not conclude that the LLM knows about the law of demand. If the responses retrieved contain a certain level of similarity with real human responses, they can serve as cost-efficient ways of reaching out to customers [14]. Though ChatGPT may simulate customer discovery and market validation activities, its accuracy must further be analyzed.

3. Methodology

As done in [14], the study centers on simulating customer WTP to the provided business model and price points. By giving ChatGPT a persona as a business analyst, it can generate a price sensitivity analysis, resembling customer discovery. As an alternative to surveys and interviews, ChatGPT's generative capability can decrease the duration of retrieving information on customer behavior. In this study, the simulations will be done using OpenAI's GPT-3.5 to determine its inherent knowledge using its base and static information.

The study conducted two types of prompting methods. The simulation utilized zero-shot inference, both with and without CoT. The prompts were applied on high-end automobile and luxury handbag businesses, both of which have illustrated conspicuous consumption in real-world markets. Getting the WTP corresponding to a certain price range can help the user compute the most optimal price point for revenue generation. Experimenting on Veblen can assist the entrepreneur in analyzing which price point can generate the most revenue whilst retaining the overall demand.

An initial zero-shot approach, without CoT, was employed wherein all the information (the business models and end-goal) were placed into a single prompt. Another method consisted of CoT, where the prompts were divided into a series of steps for the LLM to effectively process and respond to each.

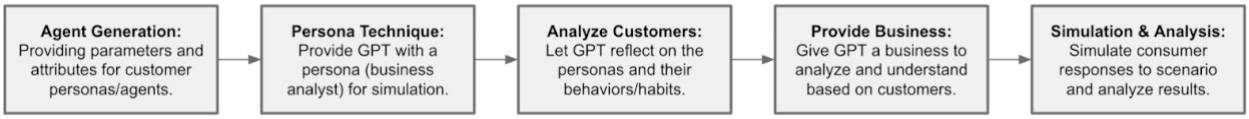


Fig. 1. Proposed prompting framework using chain-of-thought (CoT) to conduct WTP and resemble Veblen demand.

Figure 1 begins with asking GPT-3.5 to generate 30 personas from a provided list of attributes. The temperature was set to "1" in order to produce a diverse set of generated human subjects. The following personas were generated

in a JSON format, and a new set of agents was generated per trial. GPT-3.5 was then given a persona as a business research analyst. By applying this, the LLM is able to determine its role and offer preliminary insights and analysis on the generated human agents. These can be used to develop some relevance between consumers and the business model to be provided. A business model was inputted containing the characteristics of a new family business in the industries of luxury handbags and high-end automobiles. The LLM began analyzing the business model and offered perspective as to its characteristics. The insights were used by ChatGPT as a recipe for the final goal of producing a WTP curve. The CoT framework consisted of different patterns found in [16]. GPT-3.5 managed to reflect on the prompts and given information, and used its own output to produce the final goal of the experiment. Through the context, the LLM had enough time to think through clearly broken down steps. The researchers also asked the LLM to provide a rationale on the WTP curve, yielding explanations related to market preferences and perceived value. To begin the generation, the LLM was prompted to generate a randomized set of attributes for each persona with certain parameters. The LLM was prompted to generate a list of attributes associated with customer's spending behaviors, with various options provided for each attribute. Then, the LLM was tasked of randomly selecting an option for each of the attributes in this list whilst taking into considering the attribute dependencies and connections: (1) occupation, (2) age, (3) gender, (4) education level, (5) preferred shopping method, (5) interests, (6) social media usage, (7) residential area, (8) travel frequency, (9) social event participation, and (9) attitude towards sustainability.

4. Results

Two hypothetical business models [12, 26] were created to conduct the customer discovery simulation. The products were chosen specifically for the purposes of the experiment. Other industries and contexts may resemble Veblen or non-intuitive demand and normal demand curves [14].

Table 1. Business model parameters used as context for the simulation of Veblen goods.

Business	Parameters
Business model #1: Handbag business	• It mainly creates iconic handbags, crafted with exquisite material and meticulous craftsmanship. These handbags have a spacious design and allow for customizable options. • There are only a few supplies of these handbags and drops each month, which are shared through print magazines, social media platforms such as Instagram and Facebook, and on our website. • The product cannot be bought online, the customer must schedule a viewing at one of our boutiques to purchase the product.
Business model #2: Automobile business	• It primarily creates sleek cars where performance meets elegance, blending classic style and modern aerodynamics. Powered by high-performance engines that are fit for the city and winding roads. • The business releases new models of the car every year, and the drops are shared through print magazines, television advertisements, and on our website. • The product cannot be bought online, the customer must visit its stores located in urban cities to try out or test the products.

In conducting the experiment, the LLM was able to generate a WTP table on each of the given price points. The WTP results were retrieved through table format. Five conversation trials were performed for each method. Table 1 represents the number of trials that depicted a Veblen demand curve using zero-shot inference, with and without CoT prompting. This comparison aids in determining if CoT can produce more accurate simulations.

Table 2. Number of Veblen demand curves generated based on 5 trials using zero-shot inference.

Hypothetical business model	Zero-shot, *no* CoT	Zero-shot, *with* CoT
Handbag business	2	4
Automobile business	3	4

Using zero-shot learning without CoT, the LLM generated 5 out of 10 Veblen demand curves. After applying CoT, it generated 8 Veblen curves. The x-axis contains the price points of the business, and the y-axis consists of the number of average WTP from the 30 generated subjects. The line graphs refer to the average number of yeses.

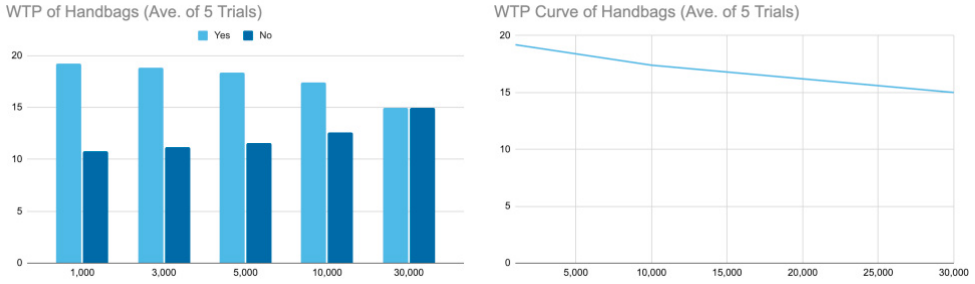


Fig. 2. WTP from the generated personas (left), and the demand curve (right) of handbags using zero-shot without CoT.

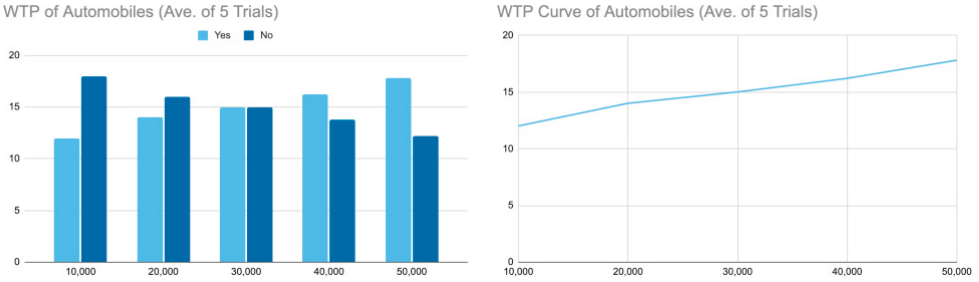


Fig. 3. WTP from the generated personas (left), and the demand curve (right) of automobiles using zero-shot without CoT.

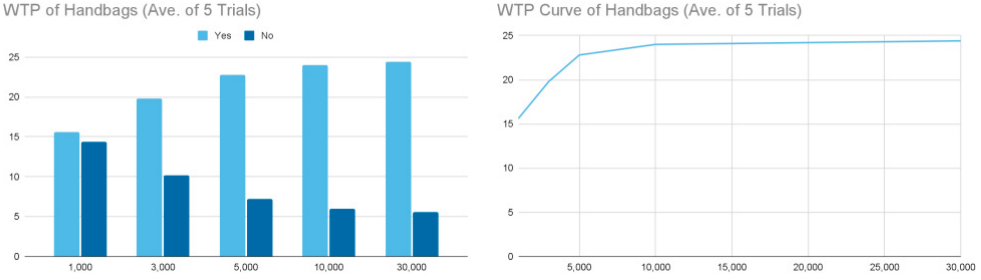


Fig. 4. WTP from the generated personas (left), and the demand curve (right) of handbags using zero-shot with CoT.

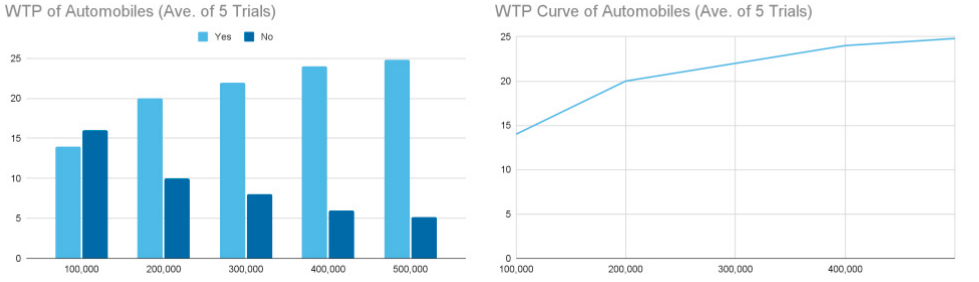


Fig. 5. WTP from the generated personas (left), and the demand curve (right) of automobiles using zero-shot without CoT.

Using the average data from the five trials, the highest price points generated the most revenue, except Figure 2. Through prompting, the results show that GPT-3.5 can generate curves that resemble Veblen demand, but this cannot conclude that the LLM has inherent knowledge of what such demand is in specific situations. The LLM also gave insights on some possible ways to iterate the business models in line with the generated customer personas. By offering perspective on better marketing and promotions, customer segmentation, in-store experience, and product modifications, ChatGPT was able to include potentially novel ideas into the model, though its value is another discussion. Prices were validated based on realistic price points in real-world businesses and markets [15]. In the context of Veblen, examining high-end goods provides a foundation for the business models inputted in the prompts. Searches from Google on "high-end handbags and automobiles" yielded links showing the following results:

- Brand X_1: \$10,400 - \$48,700 (Bags) | Brand X_2: \$1,260 - \$42,000 (Bags)
- Brand Y_1: \$243,360 - \$575,445 (Automobiles) | Brand Y_2: \$143,900 - \$330,600 (Automobiles)

Various adjustments took place in order to maintain a "clean" prompting framework. This means that nowhere in the prompt should any implications of Veblen occur. With regards to character attributes, the researchers removed any total income or monetary attributes. By removing these, the LLM refrains from using income as a signal of exclusivity, which could hinder the study from determining its inherent abilities. Interest in investments was also removed to avoid connections with conspicuous consumption and collectibles. The business models were revised to ensure that no explicit references or mentions of "luxury" were retained. As the goal was to see if GPT-3.5 could simulate the WTP of a business that resembles Veblen, this word was omitted to produce more accuracy. Both models had to be realistic, hence the characteristics were based on the perspective of a new business with limited supply due to manufacturing constraints, amateur product descriptions, and new operating procedures. Finally, the occupation choices, brand consciousness, and customer interests were revised so as not to imply the Veblen WTP. By keeping the nature of the generated human subjects more organic, the LLM can replicate real-world simulations with fairness. This can help determine ChatGPT's viability as a cost-efficient alternative for customer discovery.

5. Discussion

Revisiting the research questions presented above, together with the developed context from the simulation, yield the following results:

RQ1) What general approach may be taken to determine whether the LLM is inherently knowledgeable of Veblen products and demand, and if it can simulate customer willingness-to-pay (WTP) on certain business models?

Using GPT-3.5, zero-shot inference and the CoT technique can serve as viable simulation strategies for economic theory and WTP. However, the information presented must be realistic and clear to prevent hallucinations. It is essential to propose a business model that is reasonable and realistic, along with a range of price points that resemble real-world products. The generated agents should also resemble organic human subjects, rather than possess attributes that can imply Veblen demand. By applying various prompt patterns, including persona, context manager, reflective, recipe, and format, simulations can aid in decreasing the time it takes to perform customer discovery. GPT-3.5 contains stock knowledge of pre-trained data on economic movements, helping it generate demand that resembles non-intuitive demand curves. This does not mean that it fully understands Veblen products or non-intuitive demand, as noted in economic theory. The user must be careful in prompting as not to "beg" the simulation of Veblen curves, or else the results may be inaccurate. By integrating ABMS into manual market research, this can assist in developing a stronger foundation for future entrepreneurs entering competitive industries.

RQ2) What are the notable considerations involved in using LLMs to simulate WTP in the context of Veblen or non-intuitive demand curves?

The problem consists of non-intuitive demand curves, hence the traditional law of demand does not apply in the simulations. This produces more intricacies with regards to the perceived behavior of the generated human subjects, especially since psychological and economic factors play an important role in *conspicuous consumption*. The data quality must be analyzed. It means that the generated personas should be clear, and that the randomized attributes should be consistent with each other. For example, it sounds more reasonable for a young influencer to have a higher social media usage than others. Moreover, the LLM is still constrained by hallucinations, hence its interpretation of

the business model may be faulty at times. This is why some trials simulated normal demand curves rather than Veblen. In a more subtle note, the order of prompts must be studied since changing such order may alter the way the LLM perceives the problem. It may see the business model and personas in a broader sense, until a business analyst persona is given to it. This might change the WTP curve as well. Ethical considerations must also be discussed, as with further iterations. Given the possible use of private information in the simulations, it is imperative to implement proper safeguards that protect user data. If in the case that the prompts produced optimal accuracy of simulating customer discovery, the LLM should not hinder people from obtaining quality jobs related to market research. Moreover, due to the datasets that the LLM's were trained on, the LLM may perpetuate existing social biases that can lead to discriminatory outcomes. It is crucial not to treat the simulation as absolute representations of customer behavior. This simulation only serves as a co-pilot for entrepreneurs and managers to seek more cost-efficient and faster ways to comprehend consumer behavior and WTP.

6. Conclusion

Results show that the use of LLMs, specifically ChatGPT, can simulate customer WTP to certain price ranges. This refers to the use of prompt engineering and CoT on GPT-3.5's *homo silicus* characteristic, allowing it to replicate demand on the generated subjects. The results suggest that further iterations and studies must be performed to develop new ideas on the LLM's capability of understanding complex human behavior. Without the need to train the model, GPT-3.5 can serve as a viable co-pilot to customer discovery for future business entrepreneurs, but more experiments must be performed to improve on its accuracy, leading to lower costs.

No customized pre-training or fine-tuning was conducted as the LLM relied on its stock knowledge to generate customer discovery simulations. Moreover, the LLM was tested only with the use of zero-shot and CoT prompting, as the simulation creators focused on ChatGPT's inherent capability to detect and apply the theory of Veblen effects onto the business. Due to the nature of prompting, the lack of exposure to task-specific data may cause possible inaccuracies. Future work may include few-shot learning or fine-tuning in order for the LLM to generate more accurate and context-based customer discovery simulations. The researchers refrained from any explicit references to "Veblen goods" and the concept of non-intuitive demand curves in order to avoid helping the LLM, as the goal was to analyze its inherent knowledge on producing Veblen demand and WTP. Economic theory and "law of demand," or any reference thereof, were omitted from the prompts to avoid further implications. However, by omitting these references, there is a risk of inadequately assessing the LLM's capability to respond to the unique characteristics of Veblen goods.

Before applying prompt engineering in real-world settings, more research must be conducted on its ability to determine price point rationality, such as whether a product is deemed too expensive or cheap. Psychological human behavior can also be studied in the context of *agentic* and customer discovery simulation using ChatGPT as it can exhibit the usefulness in understanding more complex consumer behavior. This can be tested on qualitative feedback as it uses the LLM's *natural language processing* (NLP) abilities. Future simulations can also be done using GPT-4, which allows for real-time web search, which may aid in providing a better context for the given business. The idea of Veblen can be tested on other industries as well. This could create a more refined insight on non-intuitive demand curves, which can be useful for economic discussions.

Acknowledgements

We would like to thank the John Gokongwei School of Management, the Department of Quantitative Methods and Information Technology, Business Insights Laboratory for Development, and the Ateneo University Research Council of the Ateneo de Manila University for making our participation in this study and our future presentation of this study possible.

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